

Network Programming

BCA 6th Semester

1. Introduction

- Network Programming is the process of using computer code to create programs that can communicate with other programs across a network.
- This includes everything from sending and receiving data between two devices to building complex distributed systems like web servers, messaging platforms, and multiplayer games.

1.1 Features and Scope

- Word Processors like Microsoft Word open files from URLs.
- Schedule applications like Microsoft outlook automatically synchronize calendars with other employees in the company.
- IDEs like Eclipse and IntelliJ IDEA communicate with CVS (Concurrent Version System) repositories.
- Instant Messaging and Notifications.
- Online multiplayer games (Fortnite and PUBG).
- Video Streaming Services.
- Devices like thermostats, lights, and wearables communicate over the network to share sensor data or perform remote actions.

1.2 Network Programming Language, Tools & Platforms.

Different programming languages are well-suited for creating networked applications. Some of them are:

- **Python:** It is a good all-round programming language used for many different things. Python is extremely readable i.e. reading python code is relatively simple and intuitive. It has built-in support for socket programming and network protocols. Libraries like socket and asyncio make it easy to implement networking functionality.
- **Java:** A program written in Java is likely to be longer and more explicit than one written in python. Java's robust java.net package offers tools for handling sockets, HTTP, and other network protocols. It is platform-independent (write once, run everywhere), making it ideal for large, distributed network applications.

1.2 Ctd...

- **Go (Golang):** Go is designed for concurrency, making it highly efficient for handling multiple network connections simultaneously. It has built-in support and channels for managing concurrent tasks. It is used to build Scalable web servers, microservices and cloud-based applications.
- **JavaScript:** It is primarily known for its use in web development, but with Node.js, it is also used for building networked applications. Libraries like Socket.io allow real-time communication between servers and client over websockets.

1.2 Ctd...

- **C and C++** : Provides direct access to OS-level socket APIs, allowing for high performance network communication. These are used for building high-performance applications and game development.
- **Ruby**: It is high-level and easy-to-use language for network programming with libraries like EventMachine for event-driven I/O. It is used for building Web applications, APIs and chatbots.

1.2 Ctd...

There are several networking tools that assist in the development, debugging and monitoring of network applications. Here are some key tools:

- **Wireshark:** It is a network protocol analyzer used to capture and inspect network traffic. It helps developers monitor and debug network protocols in real-time. It is used for troubleshooting network issues, understanding packet flows and debugging.
- **Nmap:** It is a network scanner used for network discovery and security auditing. It aids in detecting open ports, services running on a network and performing security assessments.

1.2 Ctd...

- **Iperf3:** It is a widely used open-source tool for measuring the bandwidth performance of a network. It is typically used to test the speed between two devices connected over a network. It helps to evaluate the network's throughput and latency.
- **Cisco Packet Tracer:** It allows users to create network topologies, configure devices and simulate network traffic. It is used in network-design, configuration and testing in a virtualized environment.
- **Netstat (Network Statistics):** It is a command-line tool that displays network connections, routing tables and interface statistics. It is commonly used to monitor network activity and troubleshoot network issues.

1.2 Ctd...

- **Angry IP Scanner:** It is a fast and easy-to-use network scanner that scans IP addresses and ports. It is commonly used to discover devices on a local network and gather network information. It can scan a specific range of IP address to detect devices.
- **PingPlotter:** It is a network diagnostic tool that visualizes the performance of a network connection, primarily using ping to trace the route and packet loss in real-time. It is used for troubleshooting network latency and identifying packet loss points.

1.3 Client and Server Application

Client-server applications are programs that interact in a distributed network environment. In this mode:

- The Server is a centralized system that provides services or resources.
- The Client is a system that requests services or resources from the server.

These applications are widely used in areas such as web applications, database systems, online games and distributed computing.

How client-server Application work:

1. **Server Initialization:** The server runs on a program that listens for incoming requests on a specific port. It provides services like file access, database queries, or computations.
2. **Client Connection:** The client connects to the server using its IP address and port. It sends a request to the server to perform a specific task or retrieve a resource.
3. **Request Processing:** The server processes the client's request and sends back the response.
4. **Communication Protocol:** The interaction is based on protocols like HTTP, FTP, SMTP, or TCP/UDP.

Examples of Client-Server Applications:

- A browser like chrome requests web pages from servers (e.g. an Apache server). The server sends back the requested HTML pages.
- Whenever we use ATMs to check the bank account balance or to make a withdrawal, we actually participate in client-server communication. The ATMs are designed to interact with the back servers. The programs written on the server side serve the request sent by us through ATMs. It is obvious that ATMs act as the clients, and the bank servers act as the servers.
- Sending a mail is also an example of a client-server application. The client, who is willing to send an email, sends a connection request to the mail server. Then, the server requests authentication (password and email address) from the client. If the password and email address match, the email is sent to the destination address.

1.4 Software Design

- Software design is a mechanism to transform user requirements into some suitable form, which helps the programmer in software coding and implementation.
- It deals with representing the client's requirement, as described in SRS (Software Requirement Specification) document.
- This phase is the first step in SDLC, which moves the concentration from the problem domain to the solution domain.

Objectives of Software Design

- **Correctness:** Software design should be correct as per requirement.
- **Completeness:** The design should have all components like data structures, modules, and external interfaces.
- **Efficiency:** Resources should be used efficiently by the program.
- **Flexibility:** Able to modify on changing needs.
- **Consistency:** There should not be any inconsistency in the design.
- **Maintainability:** The design should be so simple so that it can be easily maintainable by other designers.